

HOUSE RENTAL PORTAL

A Database Management System Report

Submitted in partial fulfilment of the

Requirements for the award of the Degree of

BACHELOR OF ENGINEERING

IN

INFORMATION TECHNOLOGY

By

Vinita Mani Isukapudi (1602-22-737-062)



Department of Information Technology

Vasavi College of Engineering (Autonomous)

ACCREDITED BY NAAC WITH 'A++' GRADE

(Affiliated to Osmania University & Approved by AICTE)

Ibrahim Bagh, Hyderabad-31

2023

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Hyderabad-500031

Department of Information Technology

DECLARATION BY THE CANDIDATE

I, **Vinita Mani Isukapudi** , bearing hall ticket number, **1602-22-737-062** **hereby** declare that the project report entitled “House Rental Portal” is submitted in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering in Information Technology**.

This is a record of Bonafide work carried out by me and the results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

Vinita Mani Isukapudi

(1602-22-737-062)

(Faculty In-Charge)

(Head. Dept. of IT)

ACKNOWLEDGEMENT

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We would also like to sincerely thank the project reviewers for their valuable input and suggestions.

ABSTRACT

We present the "House Rental Portal," an innovative solution designed to simplify the process of renting residential properties. In today's dynamic real estate landscape, both tenants and landlords face challenges in finding suitable accommodations and managing rental agreements efficiently. Our portal addresses these challenges by providing a user-friendly platform that facilitates property search, selection, and booking while streamlining property management tasks for landlords.

For tenants, the portal offers a seamless browsing experience, allowing them to explore available properties based on their preferences, such as location, amenities, and budget. Detailed property listings, including high-quality images and virtual tours, empower tenants to make informed decisions without the need for physical visits. Furthermore, we provide tools for scheduling property viewings and managing lease agreements securely, enhancing the overall rental experience for tenants.

Landlords benefit from our portal's robust property management features, which enable them to create and promote property listings effectively. Through our portal, landlords can communicate with prospective tenants, schedule viewings, and manage lease agreements and rental payments efficiently. Additionally, we integrate secure authentication and encryption mechanisms to ensure the privacy and security of sensitive data.

We understand the importance of trust and transparency in the rental market, which is why our portal emphasizes clear communication and reliable information exchange between tenants and landlords. By providing transparent rental terms, including pricing details, lease agreements, and property conditions, we aim to establish a fair and mutually beneficial relationship between all parties involved.

Moreover, our commitment to innovation drives continuous improvement and adaptation to the evolving needs of the rental market. We actively solicit feedback from users and incorporate new features and enhancements to ensure that our portal remains at the forefront of the industry. Whether it's integrating advanced search filters, implementing mobile-friendly interfaces, or incorporating emerging technologies like augmented reality for virtual property tours, we strive to deliver a cutting-edge rental experience.

INTRODUCTION

The "House Rental Portal" project represents a significant advancement in the realm of real estate technology, aiming to streamline and modernize the process of renting residential properties. With the increasing complexity and competitiveness of the rental market, both tenants and landlords often encounter challenges in finding suitable accommodations and managing rental agreements efficiently. The House Rental Portal addresses these challenges by offering a comprehensive platform that revolutionizes the way properties are searched, selected, and managed.

In today's fast-paced world, convenience and accessibility are paramount. The House Rental Portal prioritizes user experience, providing tenants with an intuitive and user-friendly interface to browse through a vast array of available properties. From cozy apartments to spacious family homes, our portal caters to diverse preferences, allowing tenants to filter properties based on location, amenities, rental rates, and more. Detailed property listings, complete with high-resolution images, virtual tours, and comprehensive descriptions, empower tenants to make informed decisions from the comfort of their homes.

For landlords, the House Rental Portal offers a suite of powerful tools designed to simplify property management tasks and maximize rental potential. With just a few clicks, landlords can create compelling property listings, showcasing their properties to a wide audience of prospective tenants. Our platform facilitates seamless communication between landlords and tenants, enabling efficient scheduling of property viewings and seamless management of lease agreements and rental payments.

Key Features:

- User-friendly interface for seamless house search and selection.
- Comprehensive property listings .
- Advanced search filters based on location, amenities, and rental rates.
- Integration of secure authentication and encryption mechanisms to ensure data privacy.
- Commitment to transparency, fairness, and reliability in rental transactions.

KEY FEATURES OF PROJECT

- **Location-Based Property Selection:** Users can browse available properties based on their preferred location, facilitating targeted property search and selection.
- **Dynamic House Selection:** Interactive interface allowing users to select houses within their chosen location, providing a tailored rental experience.
- **Effortless Booking Process:** Seamless booking process for selected houses, enabling users to secure their desired accommodation with ease.
- **Real-Time Availability:** Instantaneous updates on property availability, ensuring users access up-to-date information for informed decision-making.
- **User-Friendly Interface:** Intuitive interface design for smooth navigation and enhanced user experience throughout the property search and booking journey.
- **Database Integration:** Integration with a robust database system to efficiently manage property listings, user profiles, and booking information.
- **Secure Transactions:** Implementation of secure transaction protocols to protect user data and facilitate safe and reliable rental transactions.
- **Responsive Customer Support:** Accessible customer support channels to address user queries, provide assistance, and resolve issues promptly.
- **Feedback Mechanism:** Built-in feedback mechanism for users to share their rental experiences, enabling continuous improvement of the platform based on user feedback.
- **Scalability and Adaptability:** Scalable architecture allowing for future expansion and adaptation to evolving user needs and market demands.

Technical Feasibility:

Back End

In this project I've implemented the back end of the system which is designed on "MySQL"

On this sequence query language, we created 10 tables named:

- Houses
- Customers
- Bookings
- Payments
- Reviews
- Recommendations
- CustomerPreference
- Amenities
- Houseamenities
- Housefeatures

SYSTEM REQUIREMENTS

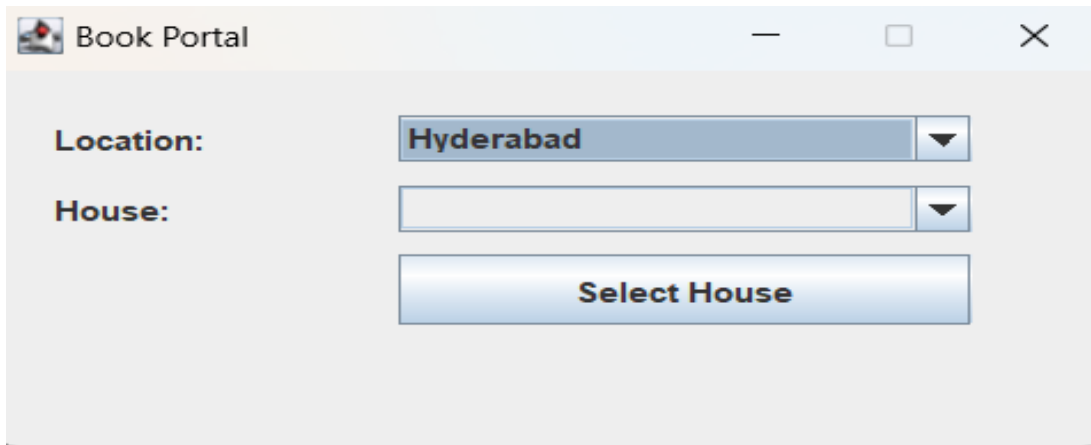
SOFTWARE REQUIREMENTS

- **Text Editor (Visual Studio Code):** Visual Studio Code is the recommended text editor for this project. It provides a lightweight, powerful environment for coding and has built-in support for HTML, CSS, JavaScript, flask, python and MYSQL.
- **Web Browser:** To test and view the project, a modern web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge is required. Ensure compatibility and responsiveness across different browsers.

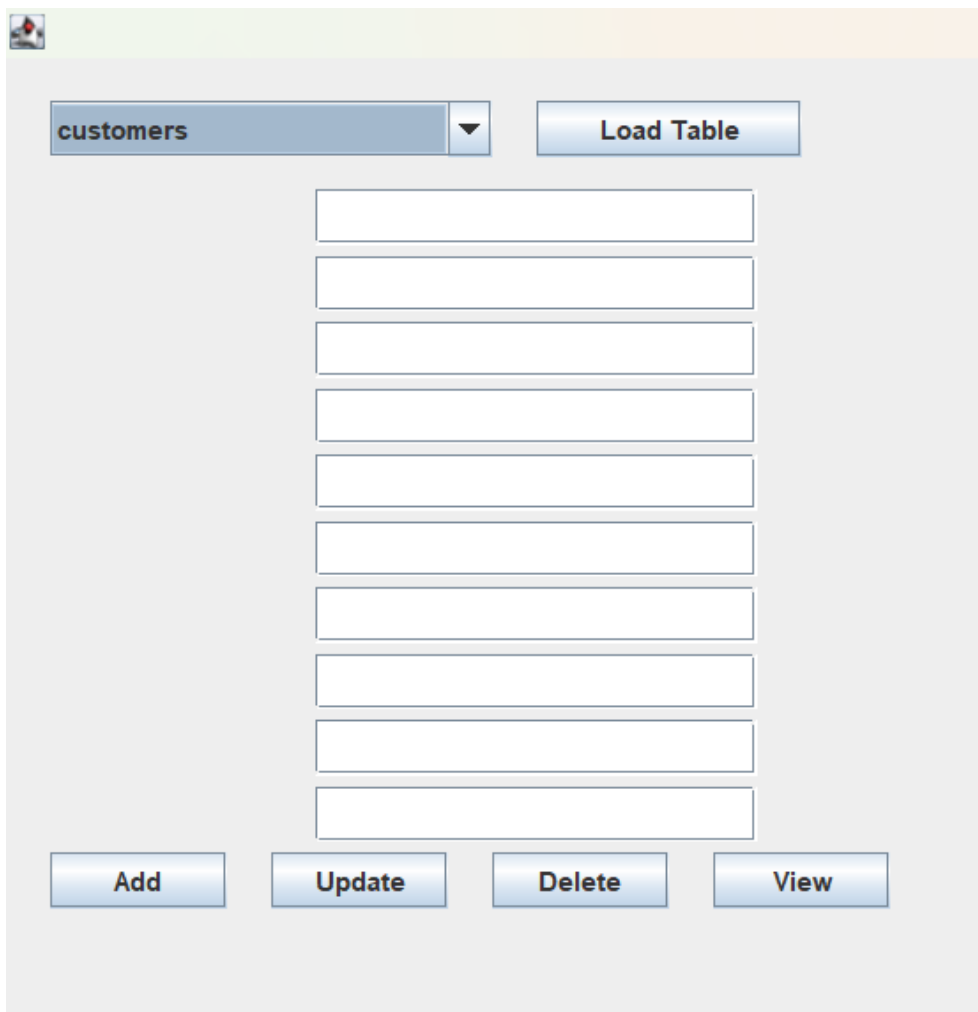
HARDWARE REQUIREMENTS

- **Processor:** A multi-core processor with a speed of at least 2.5 GHz is recommended to ensure smooth development and testing of the House Rental Portal.
- **RAM (Random Access Memory):** A minimum of 8 GB RAM is recommended to handle the demands of coding, running a local server, and testing in the web browser efficiently. This ensures optimal performance during the development and testing phases of the project.

RESULTS

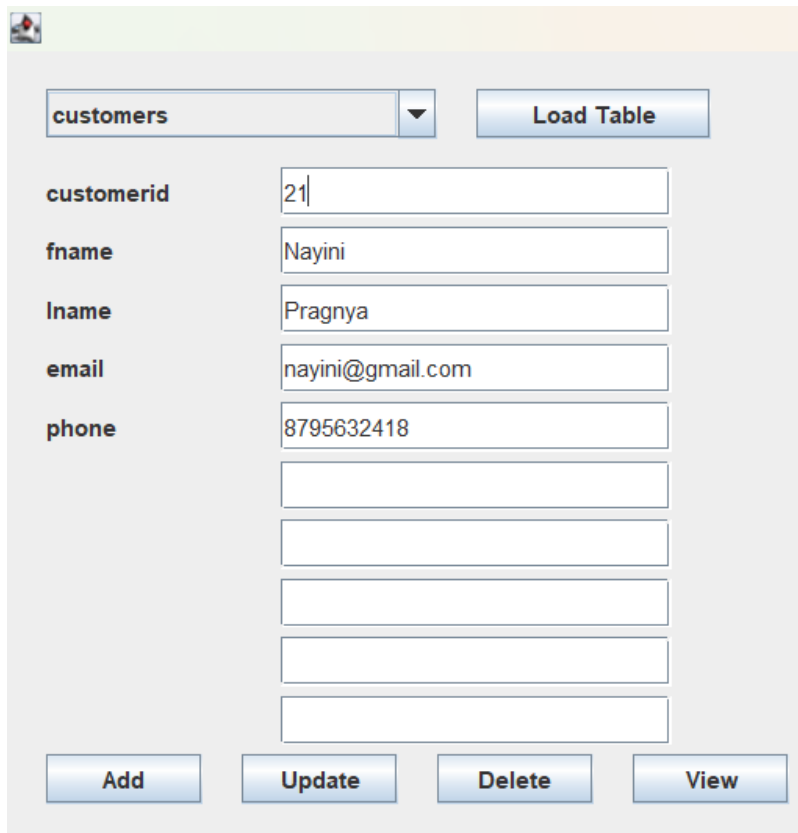


The screenshot shows a window titled "Book Portal" with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there are two labels: "Location:" and "House:". The "Location:" label is followed by a dropdown menu currently displaying "Hyderabad". The "House:" label is followed by an empty dropdown menu. Below these two dropdowns is a button labeled "Select House".

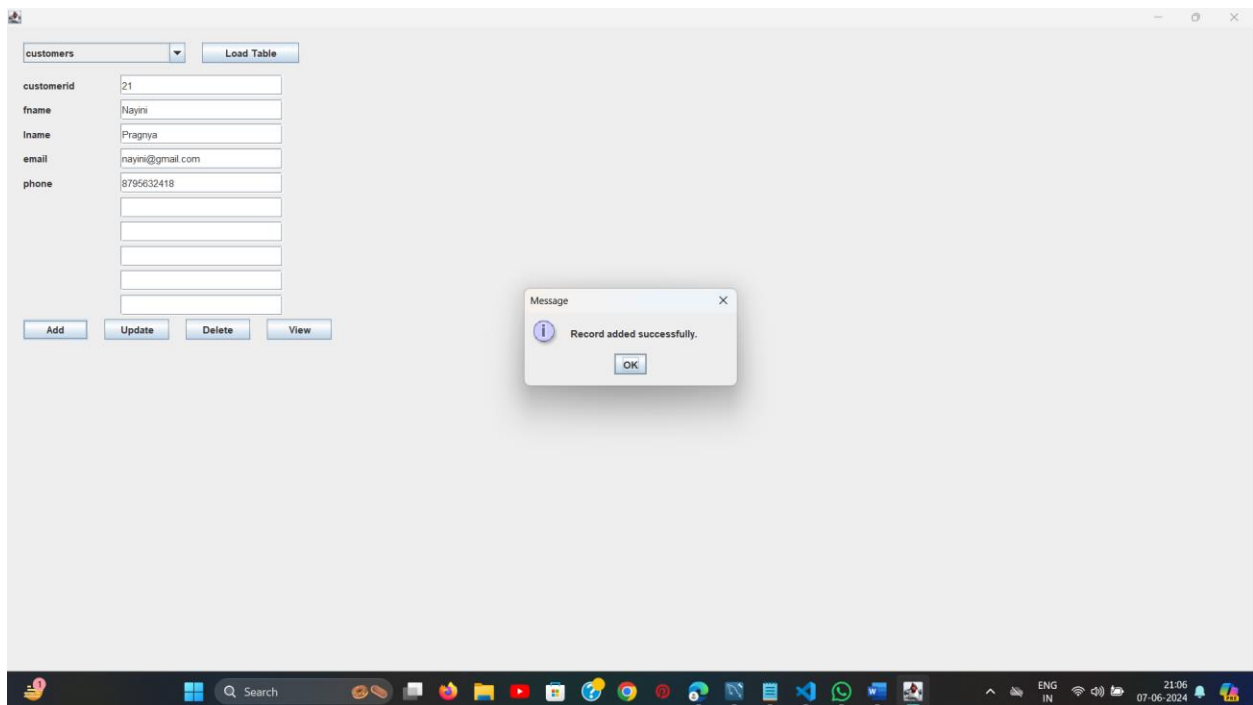


The screenshot shows the same application window with a different view. At the top left, there is a dropdown menu currently displaying "customers". To its right is a button labeled "Load Table". Below these, there is a vertical list of ten empty text input fields, likely for displaying data from a table. At the bottom of the window, there are four buttons: "Add", "Update", "Delete", and "View".

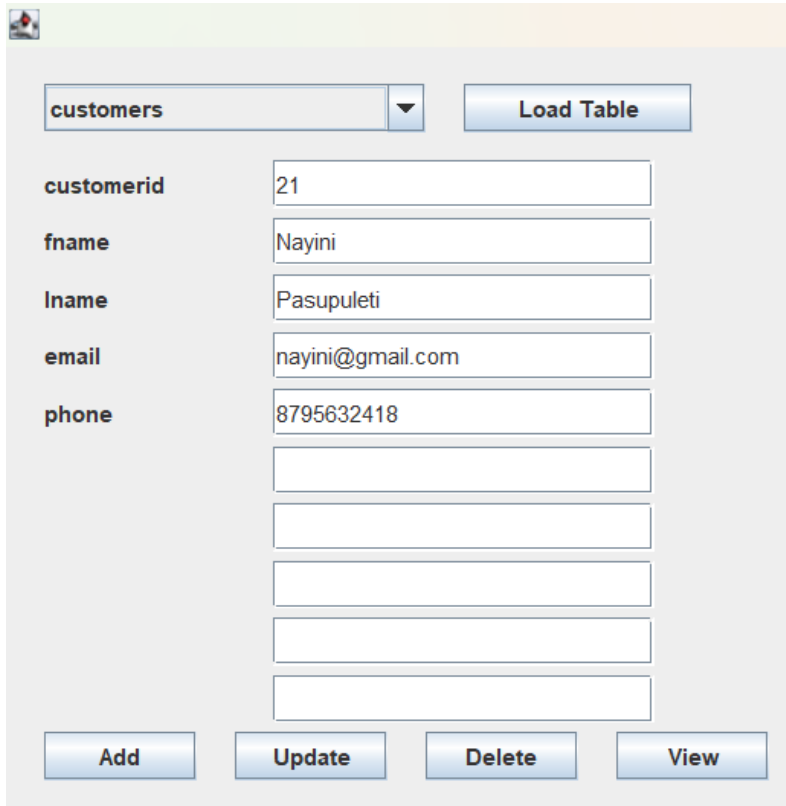
Insert



A web application interface for inserting a new customer record. At the top, there is a dropdown menu set to 'customers' and a 'Load Table' button. Below this, the form contains several input fields with labels on the left: 'customerid' (with the value '21'), 'fname' (with 'Nayini'), 'lname' (with 'Pragnya'), 'email' (with 'nayini@gmail.com'), and 'phone' (with '8795632418'). There are five additional empty input fields below the phone field. At the bottom, there are four buttons: 'Add', 'Update', 'Delete', and 'View'.



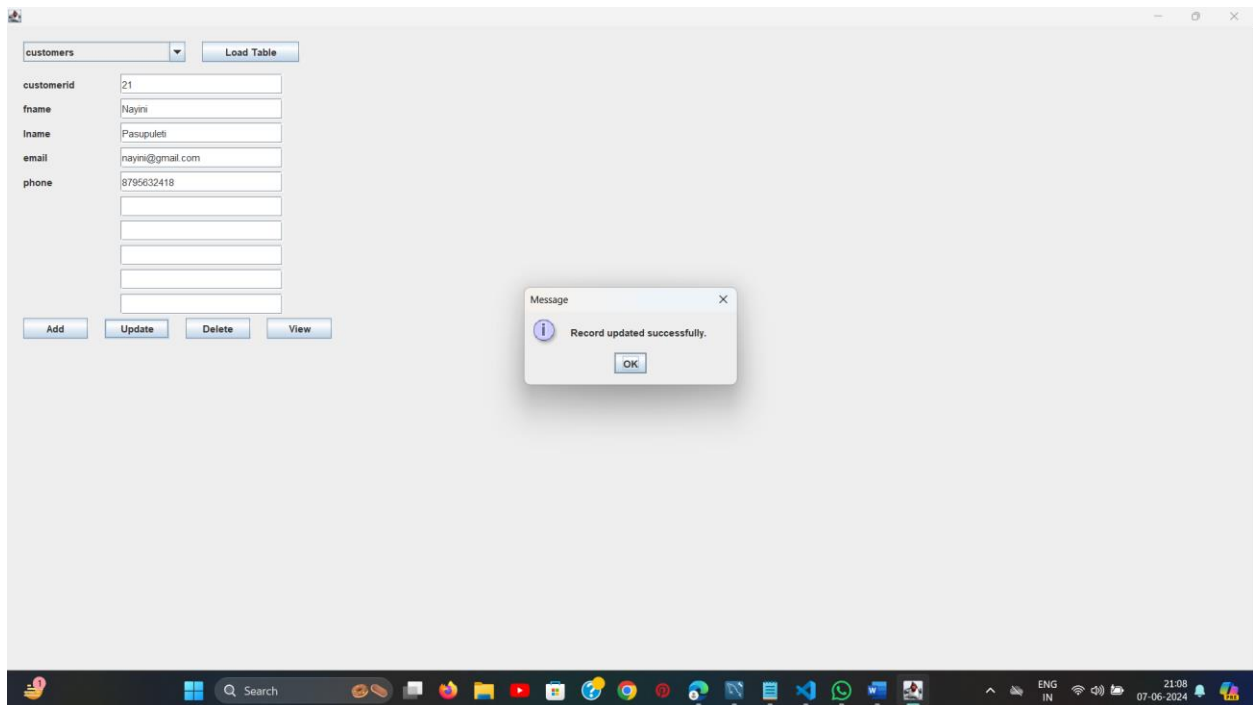
Update



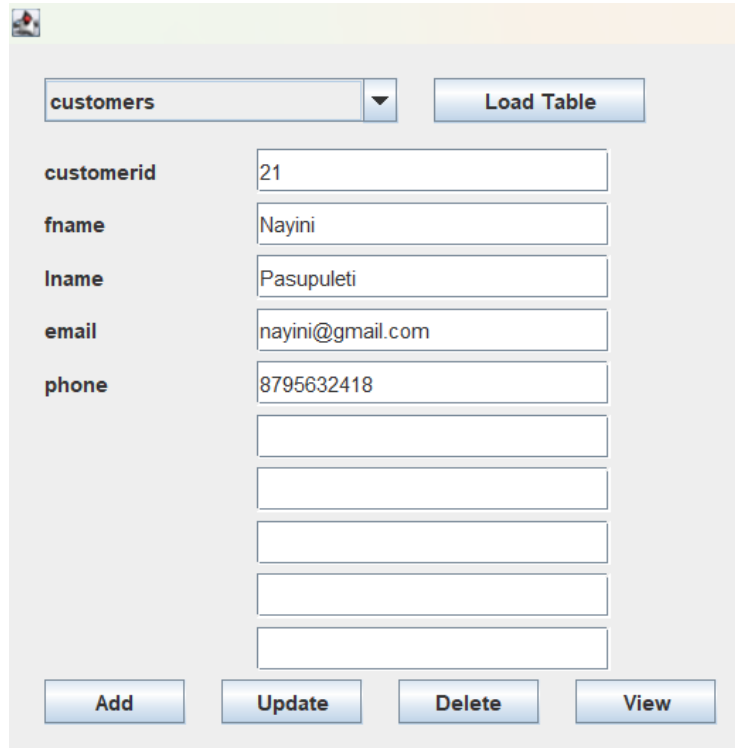
The screenshot shows a web application window with a title bar. Inside, there's a dropdown menu set to 'customers' and a 'Load Table' button. Below this, there are input fields for 'customerid' (21), 'fname' (Nayini), 'lname' (Pasupuleti), 'email' (nayini@gmail.com), and 'phone' (8795632418). There are also five empty input fields for additional data. At the bottom, there are four buttons: 'Add', 'Update', 'Delete', and 'View'.

Field	Value
customerid	21
fname	Nayini
lname	Pasupuleti
email	nayini@gmail.com
phone	8795632418

Buttons: Add, Update, Delete, View



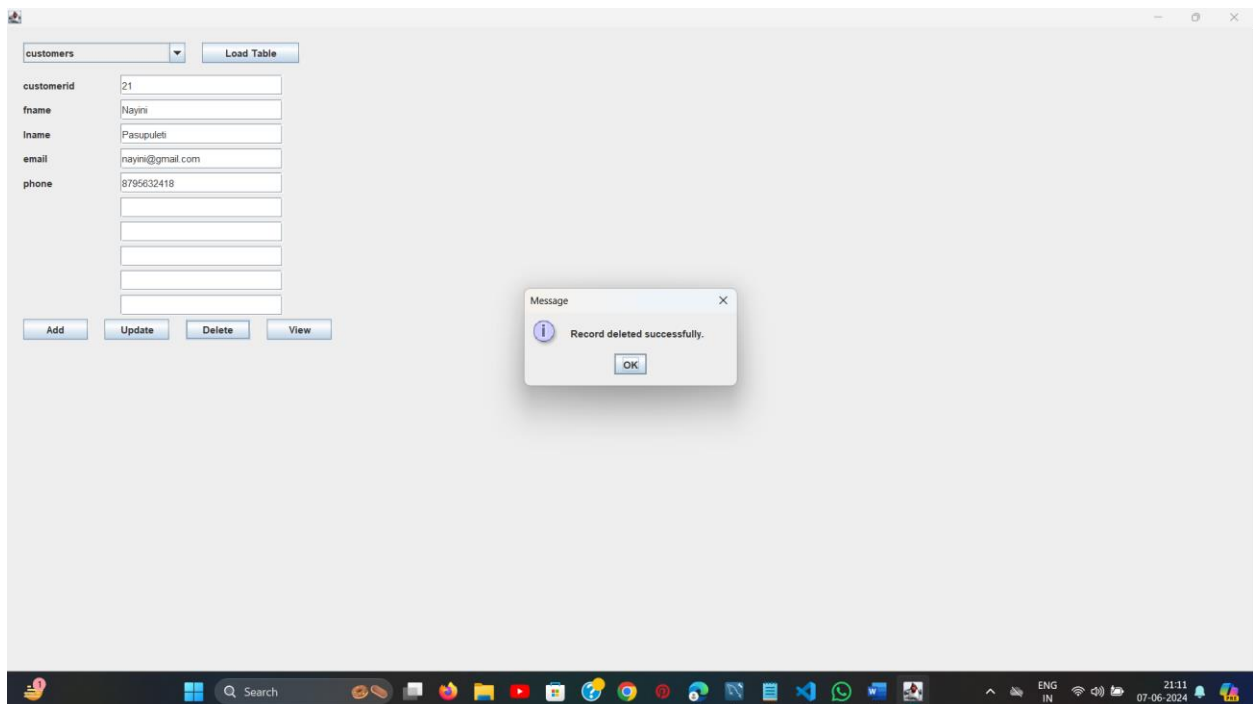
Delete



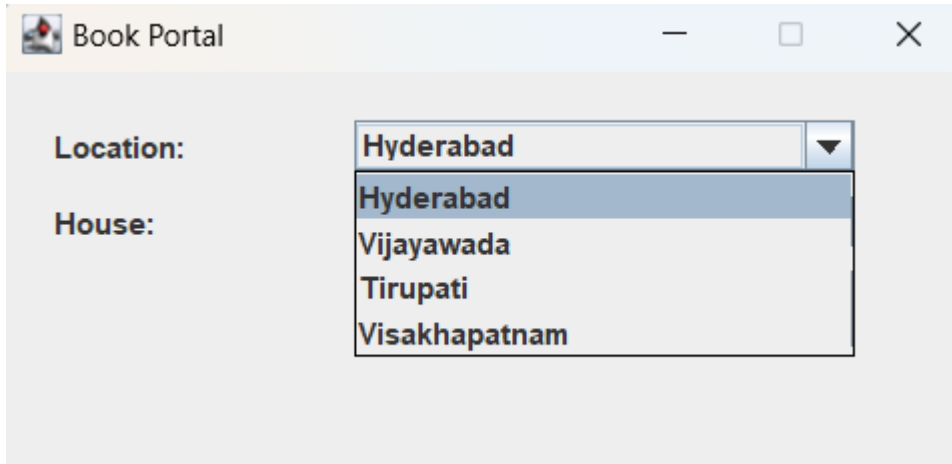
The screenshot shows a web application interface for managing customer records. At the top, there is a dropdown menu set to 'customers' and a 'Load Table' button. Below this, the details for a specific customer are displayed in a form:

customerid	21
fname	Nayini
lname	Pasupuleti
email	nayini@gmail.com
phone	8795632418

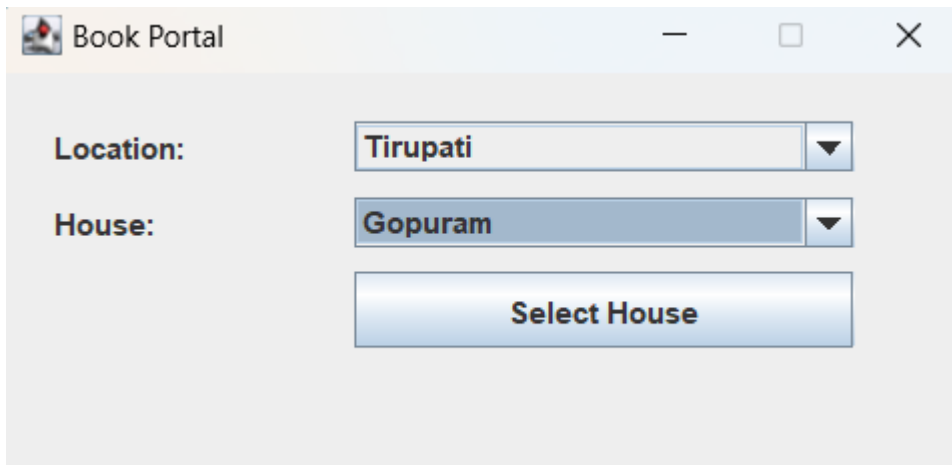
At the bottom of the form, there are four buttons: 'Add', 'Update', 'Delete', and 'View'. The 'Delete' button is highlighted, indicating it is the active operation.



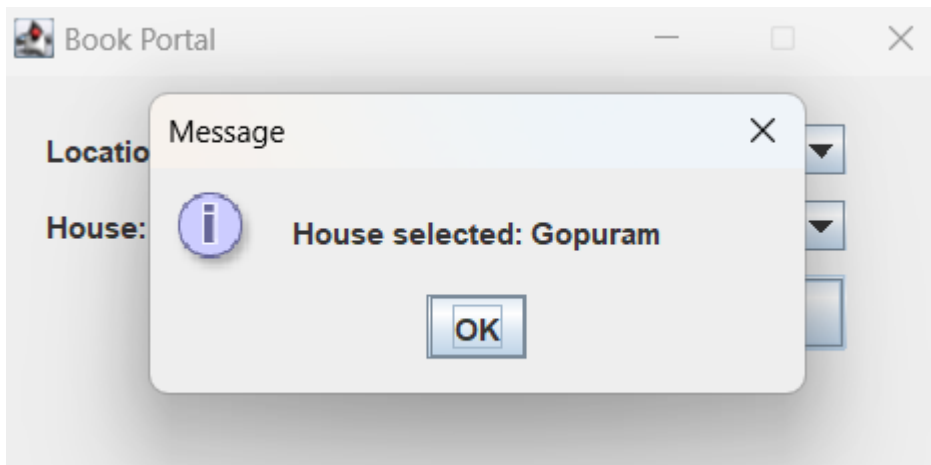
Main Portal



The screenshot shows a window titled "Book Portal" with a standard Windows title bar (minimize, maximize, close buttons). Inside the window, there are two labels: "Location:" and "House:". The "Location:" label is followed by a dropdown menu that is currently open, displaying a list of locations: "Hyderabad", "Hyderabad", "Vijayawada", "Tirupati", and "Visakhapatnam". The "House:" label is currently empty.



The screenshot shows the "Book Portal" window with the "Location:" dropdown menu set to "Tirupati" and the "House:" dropdown menu set to "Gopuram". Below these dropdowns is a button labeled "Select House".



The screenshot shows the "Book Portal" window with a message dialog box overlaid on top. The dialog box is titled "Message" and contains an information icon (a lowercase 'i' inside a circle) and the text "House selected: Gopuram". There is an "OK" button at the bottom of the dialog box. The "Location:" and "House:" labels are partially visible behind the dialog box.

 Book Portal

—

□

×

Location:

Tirupati

▼

House:

Gopuram

▼

Select House

Check-in Date (YYYY-MM-DD):

Check-out Date (YYYY-MM-DD):

Number of Guests:

Customer ID:

First Name:

Last Name:

Email:

Phone:

Book House

Book Portal

×

Location:

Tirupati

▼

House:

Gopuram

▼

Select House

Check-in Date (YYYY-MM-DD):

2024-06-15

Check-out Date (YYYY-MM-DD):

2024-06-17

Number of Guests:

4

Customer ID:

23

First Name:

Deeksha

Last Name:

Ghattamaneni

Email:

deeksha@gmail.com

Phone:

7896541253

Book House

Book Portal

×

Location:

Tirupati

▼

House:

Gopuram

▼

Select House

Message

×

i

Booking completed successfully! Total Price: \$13000.0

OK

Customer ID:

23

First Name:

Deeksha

Last Name:

Ghattamaneni

Email:

deeksha@gmail.com

Phone:

7896541253

Book House

ADVANTAGES

- ❖ **Inventory Balance:** The portal ensures a balanced inventory by accurately tracking the availability of rental properties. This helps prevent shortages and ensures that there is just enough inventory on hand without excess.
- ❖ **Inventory Turnover:** Maintaining a high turnover ratio ensures that properties are not left vacant for extended periods, maximizing profitability and preventing stagnation in the rental market.
- ❖ **Customer Retention:** By offering a seamless rental experience, the portal encourages repeat bookings from satisfied tenants, fostering long-term relationships and loyalty.
- ❖ **Accurate Planning:** Efficient inventory management allows for better forecasting and planning, ensuring that properties are available to meet demand, even during peak seasons.
- ❖ **Streamlined Operations:** The portal optimizes warehouse organization by categorizing properties based on location, amenities, and other factors, making it easier for landlords to manage their rental portfolios.
- ❖ **Improved Efficiency:** Empowering landlords with tools for inventory management streamlines operations, reduces manual effort, and maximizes resource utilization.
- ❖ **Simplified Ordering:** Landlords can efficiently manage property listings, bookings, and payments, saving time and resources through automated processes.
- ❖ **Enhanced Tracking:** For property owners with multiple listings, the portal offers centralized inventory tracking, enabling efficient management across various locations.
- ❖ **Time Savings:** Automated inventory tracking and management tools eliminate the need for manual recounts, saving time and ensuring accurate record-keeping.

CONCLUSION

House Rental Portal offers a user-friendly solution that simplifies property management for landlords and enhances the rental experience for tenants. By optimizing inventory management, streamlining operations, and leveraging technology to facilitate seamless transactions, the portal sets a new standard of efficiency and convenience in the rental market, promising increased satisfaction and profitability for all stakeholders involved.